Jayroop Ramesh jayrmh@gmail.com

I am a second year MSc. Computer Engineering at the American University of Sharjah, United Arab Emirates (UAE), and I would like to formally express my interest in joining the esteemed Oxford University as a PhD student in the Fall of 2022.

If accepted, I hope to have the privilege of working at the Department of Computer Science under the supervision of Dr. Jim Davies, Dr. Alfonso Bueno-Orovio, Dr. Ana Namburete, Dr. Blanca Rodriguez, Dr. Irina Voiculescu, or Dr. Tam Vu on topics at the intersection of artificial intelligence (AI) and healthcare. I wish to continue my research endeavours to reflect their active efforts in improving the quality of life and extending lives of patients at scale through the eclectic union of AI, medical imaging records and ubiquitously monitored low-cost wearable-derived biosignals.

I intend to specialize in public health for my graduate studies, with a focus on applying state-of-the-art machine learning methods to implement automated clinical screening systems for analysing wearable monitoring data. Specifically, I would like to study the clinical value of developing continual learning models that are able to adapt learned knowledge with the impact of genetic determinants, lifestyle factors, and comorbidities on the progression of health conditions, to possibly identify appropriate treatment interventions which can be integrated and delivered via wearables or smartphones.

I believe I can contribute to such ongoing research initiatives owing to my previous experience with developing machine learning approaches to address healthcare challenges in academia and independent projects pertaining to the domains of cardiovascular disease, sleep disorders, and depression. I possess a strong sense of familiarity with the fundamentals of data structures, algorithms, microcontrollers, embedded systems, and networking principles, repeatedly reinforced by my role as a teaching assistant at my university for two years, where I assist my professors during laboratory assignments, practice sessions, seminars, and class assessments.

Since the first year of my Master’s degree, I have become enamoured with the art and science of research, resulting in the actualization of five first-author articles international conferences and leading journals. The three that focus on healthcare are listed as follows: monitoring progression of diabetes with smartphones and glucometers [1], detecting atrial fibrillation with convolutional neural networks using wearable devices [2], and stratifying risk of sleep apnoea by leveraging ensemble machine learning trained on electronic health records [3]. For my thesis which is currently underway, I am working towards observing the ability of latent space representations derived from multi-task attention-mechanism deep learning for joint modelling of sleep apnoea and depression with sleep stage architecture and electroencephalogram band frequency measures.

I have authored two more journal articles with my academic supervisors that are currently in review. One proposes an implementation of deep transfer learning-based interpolation methods to augment medical imaging with the impact of reducing cone-beam computed tomography scanning time, while the second focuses on adapting evolutionary computation for optimizing the virtual machine allocation operations of cloud data centres. As part of two research groups at my university, I am simultaneously working on two more healthcare projects, where I am advised by an epidemiologist and an endocrinologist together with my faculty supervisors. My responsibilities are to analyse the demographic factors affecting prevalence of ocular refractive errors in school children using unsupervised machine learning

and validate the efficacy of edge-device speech-emotion classification with compressed deep learning models.

My motivation to work extensively on projects oriented towards wearable devices and electronic health records primarily stems from my desire to create prognostic utility by adding elements of artificial intelligence to existing healthcare systems and frameworks. I am a proponent of enabling accessible early screening at the population level to facilitate precision medicine, such that the right patient receives the right treatment at the right time in a fair way. During my study of these aspects, I learned to interface wearable sensors with monolithic systems through modular computing architecture, perform data mining and knowledge discovery in terms of biomedical signal processing, statistical analysis, machine learning, and deep learning, and employ metaheuristic techniques for feature selection, and model pruning. The insights garnered from my communication with doctors and health authorities have introduced me to the ethics, privacy and implicit biases that need to be evaluated before deploying methodologies in practice.

Furthermore, my ability to conceptualize and communicate ideas in a non-academic setting is exemplified by my winning of seven national AI innovation hackathons hosted by entities such as the Dubai Health Authority, Ericsson, Intel, Accenture, Etisalat, and New York University, Abu Dhabi. The latest of these culminated in an AI prototype, titled Mindspire, that uses vocal biomarkers and digital phenotypes to predict risk of clinical depression [4]. Mindspire is now awaiting approval for further study by the UAE government health regulatory body, of whose Innovation Centre I am currently working with. I showcased a prototype of this platform at GITEX 2019, a regional technology exhibition event, and will be presenting the progress again at the Dubai Expo in December 2021, as part of Ericsson Middle East Innovation Hub.

In terms of extra-curricular activities, I am a member of two three international engineering honour societies: Upsilon Pi Epsilon, Tau Beta Pi, and the IEEE Eta Kappa Nu. I was President of the Computer Club at my university during the final year of my undergraduate degree, where I organized and hosted multiple workshops and faculty-student ceremonies.

On a personal level, I am an avid reader of Victorian poetry, science fiction, epic fantasy, and philosophy. In my spare time, I pursue my literary passions with considerable ardour, and have published two short stories in my university's official magazine. I enjoy weaving stories, agnostic of the medium of delivery, and as such periodically upload educational videos synthesizing animated narratives with computer science algorithms on my YouTube channel titled Odysy Academy.

My long-term goal is to become a professor specializing in clinical AI and conduct meaningful research to provide equitable patient access to quality healthcare. I am beyond excited to learn, grow, achieve on a personal level, and contribute to the noble aspirations of Oxford University during my journey.

My Publications:

[1]https://ietresearch.onlinelibrary.wiley.com/doi/full/10.1049/htl2.12010

[2]https://www.mdpi.com/1424-8220/21/21/7233

[3]https://www.mdpi.com/2227-9032/9/11/1450

[4] https://www.ericsson.com/en/press-releases/5/2021/under-the-patronage-of-ministry-of-economy-uae-ericsson-announces-the-six-winning-teams-of-its-together-apart-hackathon